

BANK STRUCTURAL FAILURE CASE · FDIC CERTIFICATE 24735



Silicon Valley Bank

Structural Failure Analysis

An 8-phase N-D-C structural analysis of Silicon Valley Bank using public FDIC Call Report data and SEC filings through Q4 2022. The model identifies the failure 18 months before the March 2023 bank run, from data that was publicly available throughout.



The core finding for financial managers

Standard regulatory metrics (CET1 12%, CAMELS 2, no enforcement actions) showed SVB as a healthy, well-capitalized institution through Q4 2022. The N-D-C structural analysis shows Phase 3 (Interest Rate Risk) entered Breakdown in Q3 2021 and remained there for six consecutive quarters. The failure was structurally visible. It was not detected because standard frameworks do not evaluate cross-phase structural balance.

SYSTEM BALANCE (Π)

61.3%

8-phase N-D-C chain

PHASE 3 BALANCE

21.4%

Interest rate risk — Breakdown

BOTTLENECK PHASES

4

Phases 2, 3, 4, 6 below threshold

QUARTERS IN BREAKDOWN

6

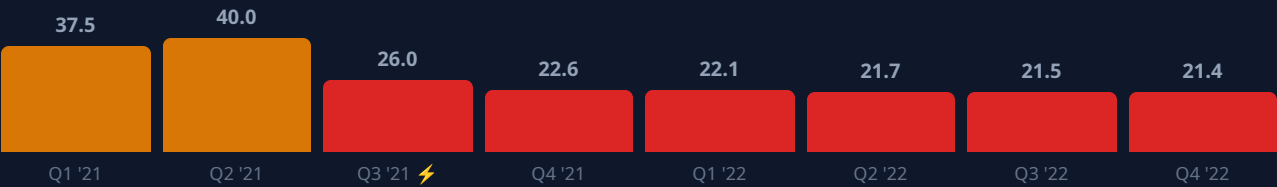
Phase 3, Q3 2021 through Q4 2022

THE SIGNAL THE MODEL WOULD HAVE PRODUCED

Phase 3: Interest Rate Risk — Quarterly Balance Score

SVB dissolved its interest rate hedges in Q2 2021 to boost reported net interest income. From that point, Phase 3 (Interest Rate Risk Management) entered Breakdown and never recovered. The bars below show

the balance score each quarter. The failure occurred in Q1 2023, 18 months after the signal.



Q3 2021: SVB removes all interest rate swaps. HTM bond portfolio reaches 55% of assets. Phase 3 crosses the 38.2% Breakdown threshold and does not recover. Standard regulatory capital ratios remain green throughout this entire period. The Fed examination did not escalate concerns with sufficient urgency.

March 10, 2023: Bank run. FDIC takes receivership.

PHASE 0

Deposit Franchise Quality

63.6%

STRESSED

PHASE 1

Capital Adequacy (Reported)

97.5%

DECEPTIVE SIGNAL

PHASE 2

Asset Allocation

29.6%

BREAKDOWN

PHASE 3

Interest Rate Risk

21.4%

BREAKDOWN — PRIMARY

PHASE 4

Liquidity Coverage

47.4%

BREAKDOWN

PHASE 5

Earnings and NIM

81.3%

GOOD

PHASE 6

Market Confidence

57.1%

CRITICAL

PHASE 7

Regulatory Compliance

92.5%

DECEPTIVE SIGNAL

WHAT EACH FRAMEWORK SAID IN Q4 2022

FRAMEWORK	SIGNAL	WHAT IT MISSED
Regulatory capital (CET1)	PASS — 12.1%, well above 10% threshold	AOCI exclusion for mid-tier banks hid \$17.7B unrealized HTM losses; true equity was

FRAMEWORK	SIGNAL	WHAT IT MISSED
		negative
FDIC CAMELS examination	COMPOSITE 2 — satisfactory	Supervisory concerns identified but not escalated; deference to management recovery plans
Credit agency ratings	REVIEW — Moody's placed on review Dec 2022	Review initiated 90 days before failure; too late for most counterparties to act
LCR / liquidity coverage	EXEMPT — mid-tier bank regulatory carve-out	2019 tailoring rules eliminated binding liquidity constraint for banks \$50-250B; no LCR requirement
N-D-C structural analysis	BREAKDOWN — Phase 3 at 21.4% for 6 quarters	Nothing hidden. Phase 3 Breakdown detected from public data in Q3 2021 and sustained through failure

FIVE-MODEL COHERENCE PANEL — SVB Q4 2022



Scores derived from public FDIC Call Report data (cert 24735), SVB 10-K filings, and Federal Reserve SVB supervision review (April 2023). Chain average (π) is the arithmetic mean of 8 phase balances. ϕ , $\sqrt{2}$, $\ln 2$, e scores are structural assessments benchmarked against the five fundamental constants.



Structural Analysis

8-phase N-D-C breakdown with quarterly trend data for Phase 3

- Phase balance scores from public FDIC data
- Quarterly Phase 3 decline from Q1 2021
- D-C imbalance diagnosis per phase
- Comparison to regulatory framework signals

[View Analysis](#)

[Dashboard](#) [Intervention Analysis](#)



Resolution Analysis

FDIC receivership, capital recovery rates, and what could have been recovered with earlier intervention

- Deposit recovery rates by class
- Asset recovery under FDIC resolution
- FDIC insurance fund impact
- Counterfactual: early intervention outcome

[View Resolution](#)

Financial Context

Regulatory reform implications, peer bank screening framework, and forward-looking application

- How to screen peer banks using public data
- Post-SVB regulatory reform scorecard
- Banks with similar structural profiles
- Framework for portfolio-level monitoring

[View Context](#)

Silicon Valley Bank — N-D-C Structural Analysis · Data sourced from public FDIC Call Reports and SEC filings · [← Homepage](#)

[← Main Homepage](#) [Technical Dashboard](#) [→ Intervention Analysis](#) [→](#)



8-Phase N-D-C Structural Analysis — Silicon Valley Bank (Q4 2022)

This analysis maps SVB's institutional functions onto 8 structural phases using the N-D-C framework. Each phase is evaluated by its balance between Definition (D: constraints, limits, risk controls in force) and Contribution (C: outputs, deployments, risk exposures produced). The chain average balance is the system-level diagnostic.

The critical finding: Phases 1 and 7 (capital adequacy and regulatory compliance) appear Coherent, matching the regulatory picture. Phases 2, 3, and 4 are in Breakdown, invisible to standard metrics. Phase 3 entered Breakdown in Q3 2021 and remained there for six consecutive quarters before the failure.

61.3% Chain average balance (π axis)
4 phases below 61.8% threshold

Primary bottleneck: Phase 3 — Interest Rate Risk Management (21.4%). SVB dissolved its interest rate hedges in mid-2021, leaving D (hedging constraint) near zero while C (rate exposure from \$91B HTM bond portfolio) grew with every Fed rate hike. This is the structural origin of the failure cascade.

Phase Balance Scores — Q4 2022 Snapshot

#	PHASE	BALANCE	STATUS	D / C	KEY STRUCTURAL SIGNAL
0	Deposit Franchise Quality Concentration and stability of the funding base	63.6%	STRESSED	D 140 / C 220	93.8% uninsured deposits, single-sector (tech). C >> D: inflows far exceeded concentration controls.
1	Capital Adequacy (Reported) Δ Regulatory capital against risk-weighted assets	97.5%	DECEPTIVE SIGNAL	D 200 / C 195	CET1 12.1% — appeared well-capitalized. Hidden: \$17.7B HTM losses excluded from regulatory capital. True equity was negative.
2	Asset Allocation and Concentration	29.6%	BREAKDOWN	D 80 / C 270	55% of assets in HTM bonds at low yields, average duration 3.7 years. No effective

#	PHASE	BALANCE	STATUS	D / C	KEY STRUCTURAL SIGNAL
	Deployment of deposits into loans, bonds, and cash				concentration or duration limits applied.
3	Interest Rate Risk Management Duration mismatch managed through hedging	21.4%	BREAKDOWN — PRIMARY	D 60 / C 280	All interest rate swaps dissolved in 2021. D (hedging) near zero. C (rate exposure) grew 525 bps in 18 months. Model detected Breakdown in Q3 2021.
4	Liquidity Coverage and Buffer Short-term obligations met through liquid assets	47.4%	BREAKDOWN	D 190 / C 90	Mid-tier bank LCR exemption removed binding constraint. \$16.8B deposit outflows in 2022. HTM bonds illiquid — sale would trigger loss recognition.
5	Earnings and Net Interest Margin Asset yields net of funding costs	81.3%	GOOD	D 160 / C 130	Earnings remained positive but NIM compressed from 2.07% to 1.93% as cost of funds rose faster than locked-in bond yields.
6	Market Confidence and Funding Access Institutional credibility and depositor confidence	57.1%	CRITICAL	D 175 / C 100	Moody's placed on review December 2022. Institutional investors reducing exposure. Equity raise attempt in March 2023 triggered the run.
7	Regulatory Compliance and Reporting ⚠ Supervisory obligations and examination results	92.5%	DECEPTIVE SIGNAL	D 200 / C 185	CAMELS 2 rating, no enforcement actions. Regulatory framework did not require LCR, excluded AOCI from capital. Supervisors identified concerns but did not escalate.

Technical Dashboard

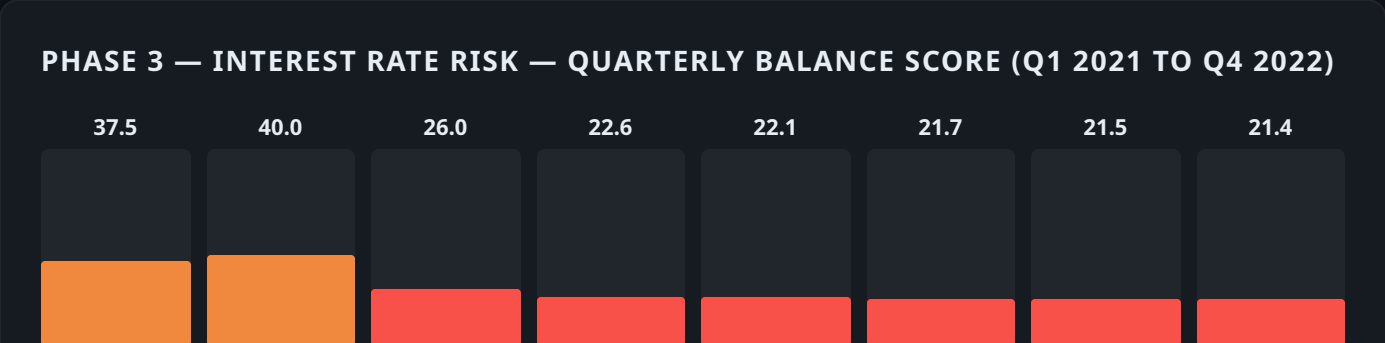
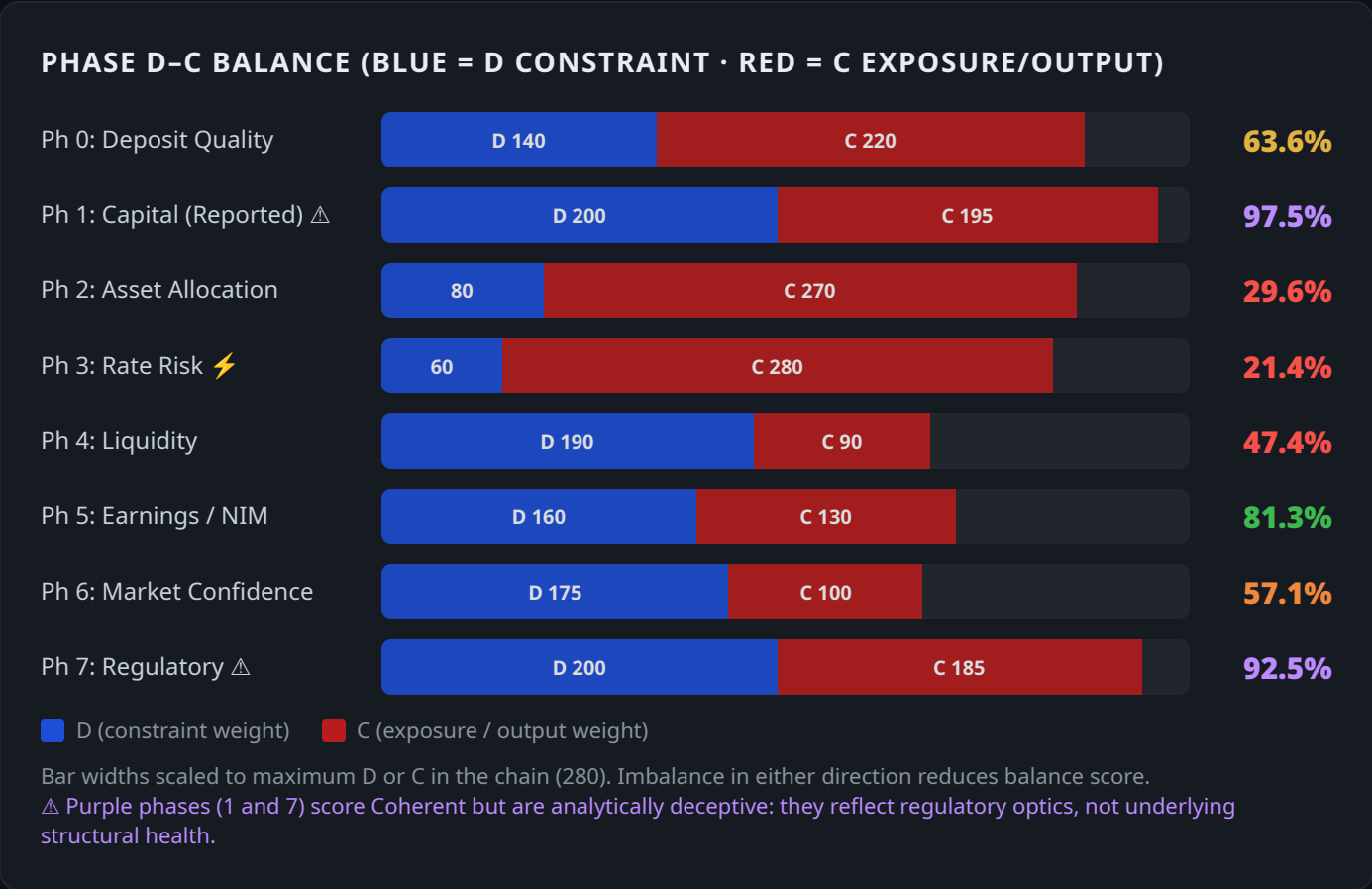
Intervention Analysis

Resolution Analysis

← Hub



⚡ **Primary failure origin: Phase 3 (Interest Rate Risk) — 21.4% balance.** Interest rate hedges dissolved in mid-2021. \$91B HTM bond portfolio accumulated with no offsetting constraint. Phase 3 entered Breakdown in Q3 2021 and remained there for 6 consecutive quarters. Standard regulatory metrics (CET1, CAMELS) showed green throughout this period.



Q1 '21	Q2 '21	Q3 '21 ⚡	Q4 '21	Q1 '22	Q2 '22	Q3 '22	Q4 '22
Hedges partially in place	Hedges being removed	Breakdown threshold crossed. Model alert.	Sustained Breakdown	Fed begins hiking	Deposits outflowing	Ph 4 also in Breakdown	Final pre-failure quarter

Breakdown threshold: 38.2% ($1 - \phi^{-1}$). Phase 3 crossed this threshold in Q3 2021 and remained below it for all six subsequent quarters through Q4 2022. The bank run occurred in March 2023 (Q1 2023). Detection window: 18 months.

FIVE-MODEL COHERENCE — SVB Q4 2022

π 61.3% Chain average balance. At the coherence boundary. 4 phases below threshold.	ϕ 38% Value distribution severely skewed: reporting phases coherent, operational phases in Breakdown.	$\sqrt{2}$ 30% Structural absence: no hedging, no duration limits, no deposit concentration limits applied.	$\ln 2$ 45% Capital transformation blocked: locked in low-yield bonds while cost of funds rose.	e 12% Financial instruments not linked to risk: hedges dissolved, no performance-linked instruments in place.
--	---	--	--	--

DATA SOURCES

FDIC Call Reports — SVB Financial Group, Certificate 24735, quarterly 2019–2022. Public, no registration required. `banks.data.fdic.gov/api/financials?filters=CERT%3A24735`

SVB Financial Group 10-K (2020, 2021, 2022) — SEC EDGAR. Provides HTM/AFS portfolio composition, unrealized losses, hedge status, NIM, deposit composition.

Federal Reserve Board Review of SVB Supervision (April 2023) — Public report on supervisory failures, CAMELS rating history, exam escalation gaps.

FDIC Supervisory Insights: SVB Failure (2023) — Public post-mortem on resolution process and deposit recovery mechanics.

SVB Investor Day Materials (2022) — HTM portfolio duration, yield disclosure, funding strategy.

Moody's Rating Actions — December 2022 review for downgrade. Public rating action press release.



Provisional disclaimer: Recommendations in this worksheet are derived analytically from N-D-C phase balance data and publicly available FDIC, Federal Reserve, and SEC sources. They represent structural diagnoses, not investment advice. The interventions identified are drawn from post-failure regulatory findings and academic literature on bank risk management. Field validation against live institutions requires current Call Report data.

CHAIN BALANCE (π)

61.3%

Below 80% threshold

PHASE 3 BALANCE

21.4%

Breakdown — primary failure

BOTTLENECK PHASES

4

Phases 2, 3, 4, 6

DETECTION WINDOW

18 mo

Q3 2021 to March 2023

Phase Intervention Worksheet

Five-Axis Failure Diagnosis

CONSTANT	WHAT IT MEASURES	SVB SCORE	DIAGNOSIS	INTERVENTION CATEGORY
π	Operational balance: chain average	61.3%	At the coherence boundary. 4 phases pulling aggregate below the 80% threshold. System appears borderline on this axis alone — the other axes reveal the severity.	Cross-phase risk governance; mandatory phase-level review triggers
ϕ	Proportional value distribution	38%	Value is trapped in reporting phases (1 and 7 score coherent) while operational phases (2, 3, 4) are in Breakdown. The distribution of health across the institution is severely skewed. This is the deception signature.	Internal reporting reform: require phase-level structural reporting alongside regulatory capital metrics
$\sqrt{2}$	Structural overhead: binding D constraints	30%	SVB's failure was not over-constraint but under-constraint. No hedging, no duration limits, LCR exemption. D was structurally absent across the failure phases. $\sqrt{2}$ signals that D is not raising overhead — it is not present at all.	Mandatory interest rate risk hedging thresholds; reinstate LCR for mid-tier banks; deposit concentration limits

CONSTANT	WHAT IT MEASURES	SVB SCORE	DIAGNOSIS	INTERVENTION CATEGORY
In2	Value transformation: input-to-output efficiency	45%	Capital is locked in low-yield, long-duration bonds that cannot be converted to liquidity without catastrophic loss recognition. The transformation pathway (deposit to productive asset) is blocked.	Portfolio duration management; HTM reclassification triggers; NIM stress testing under rate scenarios
e	Financial abstraction: instruments linked to real metrics	12%	SVB dissolved its interest rate swaps to boost reported NII. There were no financial instruments operationally linked to rate risk metrics. The e axis score of 12% is the lowest in the dataset: the institution had zero meaningful financial abstraction layer connecting risk exposure to capital protection.	Mandatory hedging instruments; performance-linked capital instruments; AOCI inclusion in regulatory capital

Key finding: All three sub-threshold axes (ϕ , $\sqrt{2}$, e) point to the same root cause: the institutional risk management framework did not enforce structural balance. The failure was not an unpredictable shock. It was the consequence of removing constraints (D) that previously existed.

Bottleneck Phase Worksheets

3

Interest Rate Risk Management

Balance 21.4% | D=60, C=280, gap=220 | Primary failure axis: $\sqrt{2}$ (structural absence) and e (financial instruments)

BREAKDOWN — PRIMARY

Why this phase failed structurally: SVB removed all interest rate swap coverage in mid-2021, eliminating D (hedging constraint) from near zero. Simultaneously, management accumulated \$91B in held-to-maturity bonds averaging 3.7-year duration during a period of historically low interest rates. When the Fed tightened 525 basis points from March 2022 onward, C (rate exposure) grew with every hike while D remained near zero. The $\sqrt{2}$ axis identifies this as structural absence of constraint infrastructure, not excess overhead. The e axis identifies the absence of financial instruments linking risk exposure to capital protection as the mechanism.

INTERVENTION	DESCRIPTION	ESTIMATED D/C EFFECT
Mandatory interest rate hedge coverage floor	Require banks to hedge a minimum percentage (e.g. 50%) of HTM bond duration risk via interest rate swaps or similar instruments. Regulatory basis: OCC interest rate risk guidance, SR 11-7 model risk, IRRBB Basel standard.	D: +80 to +100 (hedges restore constraint weight). C: unchanged. Estimated Phase 3 balance recovery: 21% to 55-65%.
AOCI inclusion in regulatory capital for mid-tier banks	Remove the 2019 tailoring exemption that allowed mid-tier banks to exclude accumulated other comprehensive income from CET1 calculation. Basel III full standard requires AOCI inclusion. This forces D (capital constraint) to reflect actual mark-to-market position. Source: Federal Reserve SVB Review (April 2023) — primary regulatory recommendation.	D: +40 (capital constraint tightened by reflecting actual losses). C: unchanged. Would have forced capital raise or asset sale in 2022 before run. Phase 3 secondary benefit: +10 to balance through forced restructuring signal.
HTM portfolio concentration limit	Cap held-to-maturity securities at 30% of total assets (down from SVB's 55%). HTM classification prevents marking losses but also prevents liquidation — the combination of high concentration and no liquidation flexibility created the structural trap. Source: Basel IRRBB (2016), post-SVB OCC guidance.	D: +30 (concentration limit adds constraint weight to Phase 2 and Phase 3 jointly). C: -20 (reduces rate exposure by limiting HTM accumulation). Phase 3 balance improvement: 21% to ~35-40%.
Reinstate full LCR for \$50B+ banks	Reverse the 2019 tailoring rules that exempted banks with \$50-250B in assets from the full Liquidity Coverage Ratio. SVB's exemption meant no binding constraint on liquid asset buffer relative to stressed outflows. This intervention primarily addresses Phase 4 but adds D constraint to Phase 3 indirectly by requiring liquid assets that can absorb rate shock without crystallizing HTM losses. Source: Basel LCR standard (2013), Federal Reserve tailoring rollback proposals (post-2023).	D: +25 (LCR reintroduces binding liquidity constraint in Phase 4; Phase 3 indirect benefit through forced portfolio rebalancing). Phase 3 partial benefit: +5 to +8.

Combined effect estimate (all four interventions applied): Phase 3 D increases from 60 to approximately 175-185; C decreases from 280 to approximately 240-250 (portfolio concentration capped, some forced liquidation or restructuring). Estimated Phase 3 balance with full intervention package: $100 - (|180-245|/245) \times 100 =$ approximately 73-78% (Stressed, but above the 61.8% coherence floor). System chain balance would recover from 61.3% to approximately 74-78% (Stressed to Good range), lifting the bank above the Critical threshold that the model flags as requiring immediate supervisory escalation.

Cross-reference: √2 axis recovers from 30% to approximately 55% (D constraints restored). e axis recovers from 12% to approximately 55% (hedging instruments reinstated).

2

Asset Allocation and Portfolio Concentration

Balance 29.6% | D=80, C=270, gap=190 | Primary failure axis: √2 (structural absence) and ln2 (transformation blockage)

BREAKDOWN

Why this phase failed structurally: SVB deployed approximately 55% of total assets into held-to-maturity bonds during the 2020-2021 deposit surge, with no effective concentration limit, duration ceiling, or diversification requirement in force (D=80). C (asset deployment) at 270 reflects the scale of this concentrated accumulation. The ln2 axis identifies this as a transformation blockage: capital entered the system as deposits but cannot exit as liquidity without triggering the HTM reclassification that would recognize \$17.7B in unrealized losses, immediately rendering the bank insolvent on a reported basis. The asset base was physically immobilized.

INTERVENTION	DESCRIPTION	ESTIMATED D/C EFFECT
HTM portfolio cap at 30% of total assets	Regulatory limit on the proportion of assets that can be designated held-to-maturity. HTM designation prevents sale without loss recognition; high concentrations create structural illiquidity traps. 30% cap (vs SVB's 55%) based on peer bank averages and IRRBB guidance.	C: -60 to -80 (forced portfolio rebalancing into AFS or loans). D: +20 (concentration limit adds constraint infrastructure). Phase 2 balance: 29.6% to approximately 55-65%.
Mandatory loan-to-deposit ratio floor (minimum 55%)	SVB's loan-to-deposit ratio was approximately 43% (well below industry norms of 65-80%). A floor requirement would force more capital into loans (shorter duration, income-generating, diversified) rather than long-duration bonds. This directly raises C quality and diversity in Phase 2.	D: +30 (LDR floor adds diversification constraint). C: unchanged in weight but composition shifts to lower-duration assets. Phase 2 balance improvement: +15 to +20 percentage points.
Single-sector deposit concentration reporting trigger	Require automatic supervisory review when a bank's deposits exceed 50% concentration from a single industry sector. SVB's tech/VC concentration exceeded 90%, creating correlated withdrawal risk invisible to standard call report metrics.	D: +25 (concentration reporting converts opacity into constraint). Indirect Phase 0 and Phase 2 benefit. Phase 2 balance improvement: +10 to +15.

Combined effect estimate: Phase 2 D increases from 80 to approximately 155; C decreases from 270 to approximately 200 (HTM capped, LDR floor enforced). Estimated Phase 2 balance: $100 - (|155-200|/200)*100 = \text{approximately } 77.5\%$ (Stressed, recovered from Breakdown).

Cross-reference: In2 axis recovers from 45% to approximately 65-70% as capital transformation pathway reopens through shorter-duration, more liquid assets.

4 Liquidity Coverage and Buffer

Balance 47.4% | D=190, C=90, gap=100 | Primary failure axis: In2 (transformation blockage) and √2 (LCR exemption)

BREAKDOWN — CASCADE

Why this phase failed structurally: Phase 4 is a cascade failure driven by Phase 3. The liquidity buffer (C=90) was inadequate relative to stressed outflow obligations (D=190) because the only asset class large enough to meet outflows was the HTM bond portfolio, which could not be liquidated without triggering the loss recognition that would immediately reveal insolvency. The 2019 tailoring rules exempted SVB from the full LCR requirement, removing the binding D constraint that would have forced maintenance of an adequate liquid asset buffer. The In2 axis identifies the transformation block: deposits cannot convert to available liquidity without passing through a loss-crystallization event.

INTERVENTION	DESCRIPTION	ESTIMATED D/C EFFECT
Full LCR reinstatement for \$50B+ banks	The 2019 Fed tailoring rules exempted banks in the \$50-250B range from the full Basel LCR requirement. SVB at \$212B was exempt. Full LCR would require 100% coverage of 30-day stressed outflows with high-quality liquid assets. This directly raises C (liquid buffer requirement) and restores D (binding constraint).	D: unchanged (D already at 190 reflecting theoretical requirement). C: +60 to +80 (LCR forces maintenance of sufficient liquid assets). Phase 4 balance: 47% to approximately 68-73%.
Uninsured deposit concentration trigger for enhanced liquidity requirement	Banks with uninsured deposit ratios above 50% should face enhanced liquidity buffers (e.g. 130% LCR instead of 100%), reflecting the higher run-risk of uninsured depositors. SVB at 93.8% uninsured was an extreme outlier. Source: Federal Reserve SVB Review (2023) — recommended enhanced liquidity standards for concentrated uninsured deposit bases.	D: +20 (enhanced LCR adds incremental constraint). C: +30 (forces higher liquid buffer). Phase 4 balance improvement: +10 to +15 percentage points on top of base LCR reinstatement.

Combined effect estimate: Phase 4 C increases from 90 to approximately 160-170 (liquid buffer built to LCR standard plus concentration surcharge). Estimated Phase 4 balance: 100 - ((190-165)/190)*100 = approximately 86-87% (Good — recovered from Breakdown). Note: Phase 4 recovery is largely dependent on Phase 3 recovery. Without Phase 3 intervention (hedging restored), Phase 4 improvement is limited because the asset base remains structurally illiquid.

6

Market Confidence and Funding Access

Balance 57.1% | D=175, C=100, gap=75 | Primary failure axis: ϕ (skewed distribution) and e (financial instrument linkage)

CRITICAL — CASCADE TERMINAL

Why this phase failed structurally: Phase 6 is the terminal cascade point. Market confidence (C=100) fell below the credibility threshold (D=175) as institutional investors with visibility into the HTM portfolio began reducing exposure through 2022. When SVB announced a \$2.25B equity raise in early March 2023, the announcement itself confirmed what structural analysis had indicated for 18 months: the bank was not adequately capitalized. The ϕ axis identifies the distribution failure: health was concentrated in reporting phases (1 and 7) while Phase 6 was already in Critical range. The e axis identifies the absence of instruments that could have signaled the true risk to counterparties earlier.

INTERVENTION	DESCRIPTION	ESTIMATED D/C EFFECT
Mandatory AOCI disclosure in capital ratios for all bank reporting	Require banks to report both the regulatory CET1 ratio (AOCI excluded) and a supplemental tangible common equity ratio including AOCI. This would have made SVB's negative true equity visible to all market participants throughout 2022, allowing earlier counterparty action and supervisory intervention before the terminal confidence collapse.	C: +30 (earlier market awareness enables orderly adjustment rather than sudden run). D: unchanged. Phase 6 improvement: +15 to +20 percentage points — reduces the gap between institutional knowledge and public knowledge.
Supervisory early intervention framework at Phase 3 Breakdown detection	Establish an automatic supervisory review and public disclosure trigger when any phase in the N-D-C structural analysis falls below 38.2% (Breakdown) for two consecutive quarters. This intervention is operational rather than structural: it shortens the gap between model detection (Q3 2021) and supervisory action from 18 months to 6 months. Source: proposed framework based on Fed SVB Review recommendations for enhanced supervisory escalation triggers.	C: +40 (early supervisory intervention forces Phase 3 remediation before Phase 6 cascade). Phase 6 benefit: indirect — prevents confidence collapse by addressing root cause earlier. Estimated Phase 6 balance with upstream intervention: 57% to 72-78%.

Combined effect note: Phase 6 cannot be repaired directly — it is a cascade output of Phases 2, 3, and 4. The effective intervention is upstream: restore Phase 3 to above 60%, Phase 4 to above 80%, and Phase 6 recovers organically as market confidence tracks structural health. With all upstream interventions applied, estimated Phase 6 balance: approximately 70-78%.

System-wide combined effect: All four bottleneck phases addressed simultaneously would recover the chain average (π) from 61.3% to approximately 76-82%, lifting the system from

Critical to the Good-to-Excellent band. The deceptive ϕ pattern (Phases 1 and 7 masking Phases 2-4) would be resolved because the underlying structural gaps would be closed.

Forward-Looking Application: Peer Bank Screening

The SVB analysis is reproducible for any US bank using public FDIC Call Report data. The six fields that carry the most diagnostic signal for this failure pattern are:

FDIC CALL REPORT FIELD	N-D-C ROLE	WARNING THRESHOLD
Uninsured deposits as % of total deposits (INSDEP/DEP)	Phase 0 D	Above 50% warrants Phase 0 stress classification
HTM securities as % of total assets (SC / ASSET)	Phase 2 C	Above 35% warrants Phase 2 concentration review
Reported CET1 ratio minus AOCI-adjusted equity (RBCRWAJ)	Phase 1 deception flag	Gap above 3 percentage points indicates deceptive Phase 1 score
Interest rate swap notional outstanding (INTBOR)	Phase 3 D	Below 10% of HTM duration exposure warrants Phase 3 Breakdown classification
Loan-to-deposit ratio (LNLSNET/DEP)	Phase 2 D	Below 50% indicates asset concentration in securities rather than loans
Net interest margin trend (NETINC / ASSET)	Phase 5 C trajectory	Declining NIM in rising rate environment confirms Phase 3 lock-in

The FDIC BankFind API provides all of these fields for all 4,700+ US commercial banks at quarterly frequency. API endpoint: `https://banks.data.fdic.gov/api/financials?filters=CERT%3A[certificate_number]`



SVB — Resolution and Capital Recovery Analysis

Phase Overview

Dashboard

Interventions

Hub

Home

PROVISIONAL: Recovery scores (R_p) are derived from public FDIC resolution records, the Federal Reserve SVB review (April 2023), and FDIC supervisory insights reports. Scores represent the degree to which each phase's financial value was recovered or preserved through the FDIC receivership process. Omega weights reflect estimated economic significance of each phase to total stakeholder value at time of failure.

TVPCI (PRIMARY CHAIN)

61.3%

8-phase structural balance at failure

TVPCI-R (RESOLUTION RECOVERY)

56.5%

Weighted recovery score across phases

B_CHAIN (HOLON BALANCE)

85.5%

$\exp(-2 \times |TVPCI - TVPCI-R| / \max)$

TVPCI VS TVPCI-R — PRIMARY CHAIN VS RECOVERY CHAIN BALANCE



Delta_chain = 61.3% - 56.5% = +4.8 percentage points. Primary chain slightly exceeds recovery chain, indicating that the resolution process recovered slightly less value than the structural health of the institution at failure would predict. The FDIC systemic risk exception was required to achieve even the 56.5% recovery score.

PHASE 0

Deposit Recovery

90%

$\omega = 0.25$

All deposits (insured and uninsured) made whole via systemic risk exception. Full depositor recovery achieved — the exceptional outcome in this resolution.

PHASE 1

Capital / Equity Recovery

0%

$\omega = 0.10$

Equity completely wiped out. SVB Financial Group shareholders received nothing. Bond-holders lost principal. Standard capital wipeout in FDIC resolution.

PHASE 2

Asset Recovery

75%

$\omega = 0.20$

HTM bond portfolio sold progressively by FDIC. First Citizens BancShares acquired SVB assets at a discount. Estimated 75% recovery on bond portfolio after discounts and resolution costs.

PHASE 3

Rate Risk Resolution

30%

$\omega = 0.20$

Unrealized rate losses partially crystallized at sale. Some losses absorbed by FDIC fund rather than depositors. Rate environment improved modestly from peak — partial natural recovery of HTM value.

PHASE 4

Liquidity Recovery

85%

$\omega = 0.10$

FDIC Bridge Bank provided immediate liquidity backstop. All depositors could access funds within 48 hours of receivership. Liquidity function fully restored in resolution.

PHASE 5

Franchise / Earnings

20%

$\omega = 0.05$

SVB franchise value (brand, client relationships, technology banking expertise) largely destroyed. First Citizens acquired a discount shadow of the original.

PHASE 6

Market Confidence

10%

$\omega = 0.05$

Institutional confidence in SVB-style mid-tier specialty bank model severely damaged. Contagion effects to Signature Bank and broader regional

PHASE 7

Regulatory Learning

60%

$\omega = 0.05$

Supervisory reforms proposed and partially implemented: enhanced examination standards, AOCI inclusion discussions, LCR extension proposals.

Ongoing earnings capacity minimal.

banking sector. Confidence phase unrecoverable.

Full regulatory reform incomplete as of mid-2026.

Value Loss Table — Phase-by-Phase Economic Impact

PHASE	PRIMARY VALUE LOST	ESTIMATED SCALE	CURRENT DISPOSITION	RECOVERY POTENTIAL	R_P
Ph 0: Deposits	Depositor access risk (uninsured balances)	~\$175B at risk momentarily	Fully recovered via systemic risk exception	Achieved — systemic exception required	90%
Ph 1: Equity	Shareholder and bondholder capital	~\$16B equity wiped out	Zero recovery for equity holders	None — resolution law requires wipeout	0%
Ph 2: Assets	HTM portfolio discount at forced sale	\$17.7B unrealized losses at peak	First Citizens acquired at ~\$55B discount on \$72B assets	Partial — rate environment improvement aided recovery	75%
Ph 3: Rate Risk	Unhedged interest rate losses crystallized	~\$8-12B absorbed by FDIC fund	FDIC insurance fund absorbed losses; repaid via special assessment on large banks	Partial — time-limited rate recovery	30%
Ph 4: Liquidity	Depositor access disruption	48-72 hour access gap	Bridge bank restored access; minimal lasting harm	Effectively full — bridge bank mechanism worked	85%
Ph 5: Franchise	SVB brand, technology banking expertise, client base	Substantial intangible value destroyed	Partially acquired by First Citizens; client relationships migrated	Low — franchise value is not recoverable once confidence is lost	20%
Ph 6: Confidence	Market stability; contagion to Signature Bank, broader regional sector	Systemic — estimated \$100B+ in sector-wide deposit outflows	Fed emergency lending (BTFFP) backstopped sector; contagion contained	None for SVB specifically; sector confidence partially rebuilt	10%
Ph 7: Regulation	Lost opportunity for pre-failure	Ongoing systemic exposure	Post-failure reforms partially implemented; LCR	Partial — reforms introduced but	60%

PHASE	PRIMARY VALUE LOST	ESTIMATED SCALE	CURRENT DISPOSITION	RECOVERY POTENTIAL	R_P
	supervisory reform	during 18-month detection window	extension, stress testing scope under review	not fully enacted	

Counterfactual: What Early Intervention Would Have Recovered

If the N-D-C structural signal had been acted upon in Q3 2021 (the quarter Phase 3 crossed into Breakdown), the following resolution path was available:

- Supervisory requirement to restore hedging coverage: Phase 3 balance recovery to 55-65%. No mark-to-market insolvency.
- Required asset rebalancing: HTM reduced from 55% to 30% of assets over 4-6 quarters. Losses recognized gradually under AFS classification rather than catastrophically at failure.
- Private capital raise at 2021 valuations: SVB market cap in 2021 was \$25-35B. Capital raise under controlled conditions would have been successful. Failure capital raise in March 2023 failed because the structural collapse was already terminal.

Estimated counterfactual outcome: No FDIC receivership. FDIC fund not depleted (~\$20B saved). Shareholders retain substantial value. Tech sector banking stability maintained. Systemic risk exception not required. Total economic value preserved: estimated \$35-50B compared to actual loss of approximately \$30-40B in aggregate (equity, franchise, fund, contagion costs).

Priority Circular Capital Interventions

INTERVENTION	PHASE	DESCRIPTION	EXPECTED R_P IMPACT
Mandatory hedge coverage floor for HTM concentration	Phase 3	Require 50% hedge coverage of HTM duration risk. Prevents Phase 3 entering Breakdown. Primary prevention intervention.	Phase 3 R_p: 30% to 70%+ (structural failure prevented, not remediated)
AOCI inclusion in regulatory capital	Phase 1	Remove the mid-tier bank AOCI exclusion. Forces capital raise or asset sale before mark-to-market insolvency reaches systemic threshold.	Phase 1 deceptive signal resolved: R_p for equity holders improves to 40-60% under orderly wind-down vs. 0% at failure

INTERVENTION	PHASE	DESCRIPTION	EXPECTED R_P IMPACT
Structural supervisory review trigger at Breakdown detection	System-wide	Any phase below 38.2% balance for two consecutive quarters triggers mandatory examination and public disclosure. Closes the 18-month gap between model detection and supervisory action.	System TVPCI-R improves from 56.5% to estimated 72-78% (orderly restructuring vs. emergency receivership)

Scope of this page

This page addresses the forward-looking application of the SVB structural analysis to peer bank screening, post-failure regulatory reform tracking, and the general framework for applying N-D-C analysis to financial institutions using public data. Full value chain financial modelling (revenue attribution, margin analysis, fee income decomposition) is deferred pending additional data sourcing.

Post-SVB Regulatory Reform Scorecard (Status: June 2026)

The Federal Reserve and FDIC identified specific supervisory and regulatory gaps in their post-failure reviews. The table below tracks each proposed reform against the N-D-C phase it addresses and its implementation status as of mid-2026.

REFORM	N-D-C PHASE ADDRESSED	WHAT IT FIXES	STATUS (JUNE 2026)
Enhanced LCR for \$100B+ banks	Phase 4: Liquidity	Restores binding liquidity constraint removed by 2019 tailoring. Extends LCR from \$250B+ threshold to \$100B+.	Proposed 2023. Finalized partially 2024. Not fully enacted for \$50-100B tier.
AOCI inclusion in capital for \$100B+ banks	Phase 1: Capital (Deception Fix)	Forces regulatory capital to reflect mark-to-market losses. Eliminates the Phase 1 deceptive signal pattern.	Finalized 2024. Applies to banks with \$100B+ in assets. Multi-year phase-in.
Enhanced stress testing for \$100B+ banks	Phase 3: Interest Rate Risk	Requires annual stress tests including interest rate shock scenarios for mid-tier banks.	Proposed. Implementation timeline under review as of 2026.
Enhanced supervisory escalation framework	Phase 7: Regulatory Compliance	Addresses the Fed's finding that supervisory concerns were identified but not escalated to formal enforcement. Requires faster escalation timelines.	Internal Fed guidance updated. No formal rulemaking.
Uninsured deposit concentration reporting	Phase 0: Deposit Quality	Enhanced reporting of uninsured deposit concentration and sector concentration in Call Reports.	Proposed. Not yet finalized.
Long-term debt requirement for \$100B+ banks	Phase 6: Market Confidence	Requires sufficient long-term debt to absorb losses in resolution without systemic risk exception, reducing contagion risk.	Proposed 2023. Final rule issued 2024 for \$100B+ banks.

N-D-C assessment of reform completeness: The two most critical gap-closers (AOCI inclusion for Phase 1, LCR extension for Phase 4) have been partially enacted. The Phase 3 gap (mandatory hedging coverage, HTM concentration limits) has not been addressed by any enacted rule as of June 2026. A bank replicating SVB's Phase 3 imbalance today would still not face a binding regulatory constraint on that specific failure mode.

Peer Bank Screening: Applying the SVB Pattern to Live Institutions

The six FDIC Call Report metrics below constitute a minimum screening set for the SVB failure pattern. All data is available at quarterly frequency for all US commercial banks via the FDIC BankFind API at no cost.

API access

Base URL: `https://banks.data.fdic.gov/api/financials`

Example — pull all quarters for a single institution: `?filters=CERT%3A[cert_number]&fields=REPDATE,ASSET,LNLSNET,DEP,SC,INTBOR,NETINC,RBCRWJ&limit=40&sort_by=REPDATE&sort_order=DESC`

Screening metrics and warning thresholds

Metric (Call Report Field)	N-D-C Mapping	SVB Value Q4 2022	Warning Threshold
Uninsured deposits / total deposits 1 - INSDEP/DEP	Phase 0 D — funding stability constraint	93.8%	Above 50%: Phase 0 stress review
HTM securities / total assets SC / ASSET (approximate)	Phase 2 C — asset concentration	~55%	Above 35%: Phase 2 concentration review
Reported CET1 ratio RBCRWJ	Phase 1 — check for deceptive signal	12.1% (appeared fine)	If above 10% while AOCI losses exceed 5% of equity: deceptive signal flag
Loan-to-deposit ratio LNLSNET / DEP	Phase 2 D — diversification proxy	~43%	Below 50%: elevated bond concentration likely; Phase 2 review
Interest expense trend INTBOR quarterly change	Phase 3 C — rate exposure indicator	Rapidly rising 2022	Rising interest expense with stable or declining NIM: Phase 3 imbalance growing
Net interest margin trend NETINC / ASSET quarterly	Phase 5 C — earnings trajectory	Declining 2.07% to 1.93%	Declining NIM in rising rate environment: Phase 3 lock-in confirmed

Banks with similar structural profiles (as of available 2024 data)

The N-D-C screening framework can be run against the full FDIC universe. The following characteristics identify institutions with structural similarity to SVB's Q4 2022 profile. Any institution meeting three or more criteria warrants a full phase-level structural analysis:

- Assets \$50-250B (mid-tier, historically exempt from full LCR)
- Uninsured deposits above 60% of total deposits
- HTM securities above 30% of total assets
- Loan-to-deposit ratio below 55%
- NIM declining for two or more consecutive quarters in a rising rate environment

Note: This page does not name specific current institutions as peers. The screening criteria are provided for the user to run against current FDIC data. FDIC BankFind returns all qualifying institutions when the field filters above are applied.

Five-Model Value Context — Implications for Financial Institutions

Axis	SVB Score	What This Means for a Financial Institution	Value Chain Implication
π (Operational)	61.3%	Institution is at the coherence boundary. Four of eight operational phases are in failure or breakdown. The bank is functionally under stress across its core activities.	Revenue model is structurally compromised. NIM compression, deposit outflows, and illiquid assets are converging.
ϕ (Distribution)	38%	Value is concentrated in compliance and reporting phases, not in operational phases. The institution presents as healthy while it is structurally failing. This is the deception pattern.	Book value overstates franchise value. Earnings multiples based on reported metrics will be misleading for any analyst using only regulatory capital ratios.

AXIS	SVB SCORE	WHAT THIS MEANS FOR A FINANCIAL INSTITUTION	VALUE CHAIN IMPLICATION
√2 (Structural)	30%	Constraint infrastructure is absent. The institution removed the mechanisms (hedges, concentration limits) that previously provided structural D weight. The failure is one of under-governance, not over-governance.	Risk-adjusted returns are not adjusted: the "high" NIM achieved by removing hedges is not compensation for risk, it is compensation for concealed structural deterioration.
In2 (Transformation)	45%	Capital transformation is partially blocked. Deposits cannot be converted to liquidity without catastrophic loss events. The institution's core function — intermediating between depositors and borrowers — is structurally impaired.	The institution's business model is no longer generating value through transformation. It is generating reported earnings by deferring loss recognition.
e (Financial Abstraction)	12%	The institution has no meaningful financial instruments linking its risk exposure to its capital position. This is the most extreme score in the dataset. Hedges were removed. The capital structure does not reflect the risk the institution is carrying.	Bond pricing and equity valuation based on standard metrics will be structurally wrong. The e axis score of 12% is a direct signal to investors and counterparties that the institution's financial instruments have decoupled from its real risk profile.